

Chapter: 15

KINETIC THEORY OF GASSES

TEMPERATURE:

TEMPERATURE:

Definition:

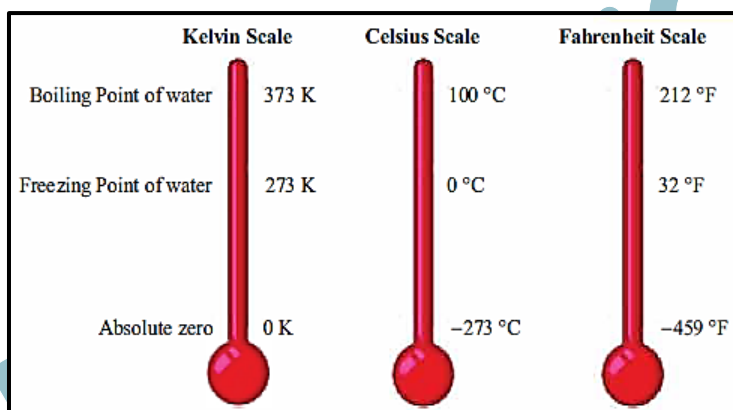
"The measure of the average translational kinetic energy of the molecules of body is called temperature of that body"

SCALES OF THERMOMETER:

1. Celsius or Centigrade Scale

2. Fahrenheit Scale

3. Kelvin or Absolute Scale.

**1. CELSIUS OR CENTIGRADE SCALE (ANDERS CELSIUS)**

In the Celsius scale, the freezing point of water or Melting point of ice is marked 0°C and boiling point is 100°C. The interval between these two points is divided into hundred equal parts. Each part represents one degree Celsius (1 °C).

2. FAHRENHEIT SCALE (DANIEL FAHRENHEIT):

In Fahrenheit scale, the fixed points are marked 32°F and 212°F respectively and the interval between the fixed points is divided into 180 equal parts. Each part represents 1 °F.

3. KELVIN SCALE (JAMES LORD KELVIN):

In this scale the melting point of ice is 273K and the boiling point of water is 373K. The temperature is given in units called Kelvin instead of degrees. The lowest temperature is 0K known as absolute zero.

RELATION BETWEEN SCALES OF TEMPERATURE:

To derive empirical formulae among Centigrade, Fahrenheit and Kelvin scales, let the three thermometers be placed in a bathtub and the mercury in each thermometer rises to the same level as shown in figure.

The relation between these scales is:

$$\frac{\text{Temperature on one Scale}-F.P.}{B.P.-F.P.} = \frac{\text{Temperature on other Scale}-F.P.}{B.P.-F.P.}$$

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} = \frac{T_K - 273}{373 - 273}$$

$$\frac{T_C}{100} = \frac{T_F - 32}{180} = \frac{T_K - 273}{100}$$

$$\frac{T_C}{5} = \frac{T_F - 32}{9} = \frac{T_K - 273}{5}$$

WORKED EXAMPLE 15.1

The normal human body temperature is 36.88°C . What is this temperature on Fahrenheit Scale?

SELF-ASSESSMENT QUESTIONS:

- Convert each of the following temperatures from the centigrade scale to Kelvin scale and Fahrenheit.
 0°C , 20°C , 120°C , 500°C , -23°C , 200°C .
- Convert each of the following temperatures from the Kelvin scale to the Celsius scale and Fahrenheit.
 0K , 20K , 100K , 300K , 373K , and 500K .

THE TRIPLE POINT OF WATER:

The vapor, liquid, and solid states of water can coexist in thermal equilibrium only at a specific pressure and temperature.

The triple point of water occurs at a specific temperature and pressure, where all three phases are in thermodynamic equilibrium.

This means that:

At the triple point, ice, liquid water, and water vapor can exist together without any phase changing to another.

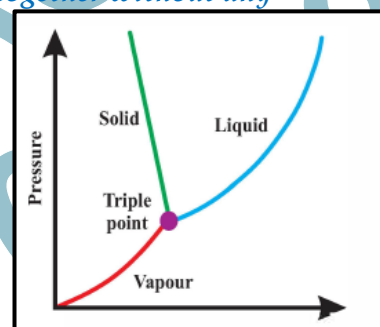
These conditions are used to define the Kelvin temperature scale, where the temperature at the triple point of water is precisely 273.16 K (0.01°C) and 4.58 mm of mercury (611.73 Pa) and is used to calibrate thermometer.

In figure, the triple point is represented as the point where the lines separating the solid, liquid, and gas phases meet.

This diagram visually demonstrates the relationship between temperature and pressure for the different phases of water.

The temperature at which Triple point of water has been set by international agreement to be $T_3 = 273.16\text{ K}$.

In T_3 the subscript 3 means "Triple Point" this agreement also sets the size of the Kelvin as $\frac{1}{273.16}$ of the difference between the Triple point temperature of water and absolute zero.

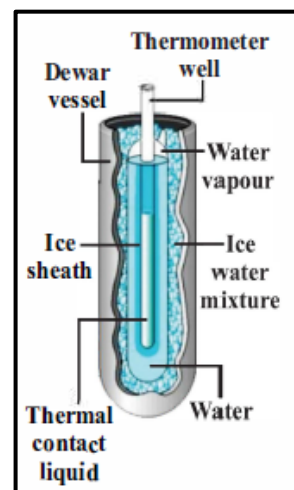
**TRIPLE POINT CELL:**

A cell in which solid ice, liquid water and water vapours co-exist in thermal equilibrium is called a triple point cell.

The bulb of the constant volume gas thermometer is shown inserted into the well of the cell.

SELF-ASSESSMENT QUESTIONS:

- What are the conditions (temperature and pressure) at which the triple point of water occurs?
- Where is the triple point located in relation to the solid, liquid, and gas regions?

**GAS LAWS:**

Gases have no fixed volume or shape and their volume at the same mass can be altered by changing the pressure as well as the temperature.

Gases can be described with the help of four variables i.e. pressure, volume, mass and temperature. The relation between any two variables is found experimentally while keeping the other two constant.

1. BOLYE'S LAW:**Definition:**

"Volume of a given mass of a gas is inversely proportional to the pressure, at constant temperature."

Mathematical Representation:

We can write Boyle's law mathematically as follows:

$$P \propto \frac{1}{V}$$

Or

$$P = \frac{K}{V}$$

Or

$$PV = K \dots \dots (1) \text{ (K = Constant)}$$

Thus, under constant temperature, if P_1 be the pressure of a gas, V_1 be its volume then:

$$P_1 V_1 = K \dots \dots (2)$$

If the pressure is changed to P_2 then the volume becomes V_2 , so that,

$$P_2 V_2 = K \dots \dots (3)$$

From equation (2) and (3) we get:

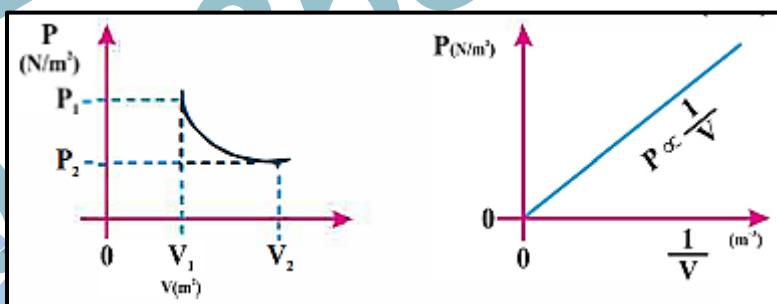
$$P_1 V_1 = P_2 V_2$$

Other Definition:

"The product of pressure and volume of a given mass of a gas remains constant, at constant temperature".

Graphical Representation:

1. The graph plotted between P and V at constant temperature is a curve called hyperbola showing the inverse relation between them for two different states, while
2. The graph of P plotted against $1/V$ is a straight line passing through the origin, showing direct proportionality.

**2. CHARLE'S LAW:****Definition:**

"The volume of given mass of a gas is directly proportional to absolute temperature at constant pressure."

Mathematical Representation:

$$V \propto T$$

Or

$$V = K T$$

Or

$$\frac{V}{T} = K \text{ (K = Constant)}$$

It means that the ratio of volume and absolute temperature of a given mass of gas at constant pressure is constant.

If V_1 and V_2 were volume of a given mass of a gas at T_1 K and T_2 K temperature, under constant pressure then, we get,

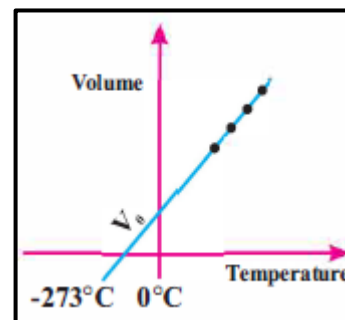
$$\frac{V_1}{T_1} = K \dots\dots (1)$$

And

$$\frac{V_2}{T_2} = K \dots\dots (2)$$

Equating the equation (1) and (2), we get,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



Graphical Representation:

The graph between volume and temperature is a straight line.

If the graph is extrapolated backward, it cuts the temperature axis at -273°C . This graph shows that the volume of a gas is zero. Kevin selected this temperature (-273°C) as the zero called absolute zero (0K).

3. AVOGADRO'S LAW:

In 1811, Italian scientist Amedeo Avogadro suggested his hypothesis regarding the relationship between volume and number of molecules of a gas. This hypothesis is now called Avogadro's law.

Statement:

"Equal volume of all gases contains the same number of molecules at the same temperature and pressure".

Mathematically:

The volume (V) of a gas is directly proportional to the number of moles (n) of the gas at constant temperature and pressure.

In symbol, we can write as: $\frac{V}{n} = K$ (at constant temperature and pressure)

Where "V" is volume and "n" is the number of moles of a gas.

Therefore 1 dm^3 (or cm^3 , m^3) of oxygen contains the same (n) number of molecules as 1 dm^3 or 1 cm^3 etc., of hydrogen or of any other gas, provided the volumes are measured under the same conditions of temperature and pressure.

It can be written as: $\frac{V_1}{n_1} = \frac{V_2}{n_2}$

Where " V_1 " and " V_2 " are volumes of gas and " n_1 " and " n_2 " are amount of gas.

4. GENERAL GAS LAW:

Statement:

"An interrelation among the physical quantities e.g. pressure, volume, temperature and amount of matter of a given sample of gas is termed as "equation of state" of gas or General gas law". OR

"The product of pressure and volume of a given mass of a gas is directly proportional to the number of moles and absolute temperature of gas." i.e.

$$P V \propto n \cdot T$$

DERIVATION OF GENERAL GAS LAW:

If the volume, pressure, and temperature of a given mass of gas are all changing, then Boyle's law and Charles's law are applicable at the same time.

To derive general gas law, we make use of Boyle's law, Charles's law and Avogadro's law.

According to Boyle's law:

"The volume of a given mass of a gas is inversely proportional to its pressure, if temperature remains constant i.e.

$$V \propto \frac{1}{P} \quad (\text{when } n \text{ number of mole and temperature } T \text{ are kept constant})$$

According to Charles's law:

"The volume of a given mass of a gas is directly proportional to absolute temperature, if pressure remains constant." i.e.

$$V \propto T \quad (\text{when } n \text{ number of mole and pressure } P \text{ are kept constant})$$

According to Avogadro's law:

"The volume (V) of a gas is directly proportional to the number of moles (n) of the gas at constant temperature and pressure." i.e.

$$V \propto n \quad (\text{when Temperature } T \text{ and pressure } P \text{ are kept constant})$$

On combining the above laws, we can get,

$$V \propto \frac{n.T}{P}$$

$$V = \frac{R.n.T}{P}$$

Where, "R" is constant of proportionality and is called General gas constant or universal gas constant and does not depend on the quantity of gas in the sample.

If P is measured in Nm^{-2} , V in m^3 and T in Kelvin, then the value of universal gas constant is $R = 8.314 \text{ J mol}^{-1} \text{K}^{-1}$

The above equation is written as: $PV = nRT$

WORKED EXAMPLES 15.2:

A cylinder contains 0.80 dm^3 of nitrogen gas at a pressure of 1.2 atmosphere ($1 \text{ atm} = 1.01 \times 10^5 \text{ N/m}^2$). A piston slowly starts compressing the gas to a pressure of 6.0 atm. The temperature of the gas remains constant. Calculate the final volume of the gas. (Temperature of gas is constant, and its mass is fixed).

WORKED EXAMPLE 15.3:

Find the mass of the air within the car tyre, given that it holds 0.020 m^3 of air at 27°C and a pressure of $3.0 \times 10^5 \text{ N/m}^2$. The molecular mass of air is 28.8 g/mol .

SELF-ASSESSMENT QUESTIONS:

1. How does the pressure of a gas change if its volume is halved at constant temperature?
2. What happens to the volume of a gas when its temperature increases while keeping the pressure constant?
3. What is the relationship between the volume and the number of moles of a gas at constant temperature and pressure?

KINETIC THEORY OF GASES:

Kinetic theory of gases explains the microscopic properties of gases in terms of the interaction of gas molecules and the energy they possess.

This theory is based on the following assumptions:

BASIC POSTULATES OF KINETIC THEORY OF GASES:

1. Gas contains a very large number of particles called molecules. Depending on the gases, each molecule consists of an atom or a group of atoms.

2. A finite volume of a gas consists of a very large number of molecules. This assumption is justified by experiments. At standard conditions there are 3×10^{25} molecules per cubic meter.
3. The size of the molecules is much smaller than separation between molecules; it is about 3×10^{-10} m.
4. The molecules move in all directions at various speeds and collide elastically with one another and with the walls of the container.
5. The molecules exert no force on one another except during collisions. In the absence of the external forces, they move freely in straight lines.
6. Laws of mechanics are assumed to be applicable to the motion of molecules.

PRESSURE OF GAS:

The pressure exerted by gas is merely the momentum transferred to the walls of the container per second per unit area due to the continuous collisions of molecules of the gas.

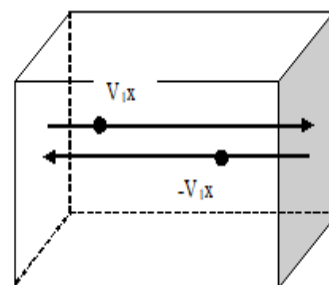
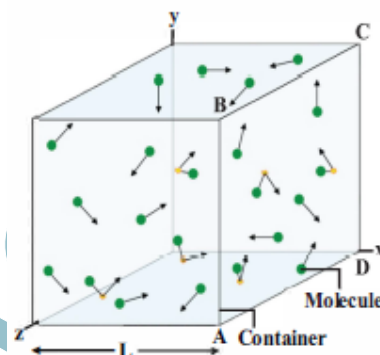
To calculate the pressure of an ideal gas from Kinetic Theory. Let us consider a cube having side length L whose walls are perfectly elastic containing N number of molecules each of mass m as shown in figure.

Consider a single molecule of mass " m " moving with velocity " V_1 " parallel to x -axis. It moves back and forth, colliding at regular intervals with the ends of the box and thereby contributing to the pressure of the gas.

A molecule which has a velocity " V_1 " can be resolved into three rectangular components V_{1x} , V_{1y} and V_{1z} parallel to three co-ordinates axis x , y and z .

A molecule which collides with the face ABCD of the cube, will rebound elastically in opposite direction, such that x -component of the velocity V_{1x} , is reversed, the V_{1y} and V_{1z} remain unaffected.

Therefore, the momentum before collision is mV_{1x} and after collision is $-mV_{1x}$ causing a change of momentum.



$$\begin{aligned}
 \text{Momentum before collision} &= m v_{1x} \\
 \text{Momentum after collision} &= -m v_{1x} \\
 \text{Change in momentum; } \Delta p &= m v_{1x} - (-m v_{1x}) \\
 &= m v_{1x} + m v_{1x} \\
 \Delta p &= 2 m v_{1x} \dots\dots\dots (1)
 \end{aligned}$$

After recoiling, the molecule travels to the opposite face and collides with it, rebounds and travels back to the face ABCD after covering a distance $2L$. The time Δt between two successive collisions with face ABCD is:

$$\begin{aligned}
 v_{1x} &= \frac{2L}{\Delta t} \\
 \text{Or } \Delta t &= \frac{2L}{v_{1x}} \dots\dots\dots (2)
 \end{aligned}$$

Now we can find the force that this one molecule exerts on the face ABCD, using Newton's 2nd law of motion.

This says that the rate of change of momentum of the molecule is equal to force applied by the wall.

According to Newton's 3rd law of motion, force F_{1x} exerted the molecule on face ABCD is equal but opposite.

$$\begin{aligned}
 \text{Rate of change of momentum} &= \frac{\text{Change of momentum}}{\text{Time taken}} \\
 F_{1x} &= \frac{\Delta p}{\Delta t} \dots\dots\dots (3)
 \end{aligned}$$

By putting the values from equations (1) and (2) in equation (3), we get

$$\begin{aligned}
 \text{Rate of change of momentum} &= \frac{2m v_{1x}}{\frac{2L}{v_{1x}}} \\
 F_{1x} &= \frac{m v_{1x}^2}{L}
 \end{aligned}$$

Since, rate of change of momentum is called as a force, therefore the force exerted by a single molecule on the wall of cube along x-axis is

$$F_{1x} = \frac{mv_{1x}^2}{L}$$

Similarly, force exerted by all molecules ($\frac{1}{3} N$ molecules) along x-axis on the yz-plane will be.

$$F_{2x} = \frac{mv_{2x}^2}{L}$$

Force by second molecule

$$F_{3x} = \frac{mv_{3x}^2}{L}$$

Force by third molecule and similarly

And so on:

$$F_{nx} = \frac{mv_{nx}^2}{L}$$

Similarly, the forces due to all other molecules can be determined. Thus, the total x-directed F_x due to N number of molecules of the gas moving with velocities $V_{1x}, V_{2x}, V_{3x}, \dots, V_{nx}$ is:

$$\begin{aligned} F_x &= F_{1x} + F_{2x} + F_{3x} + \dots + F_{nx} \\ F_x &= \frac{mv_{1x}^2}{L} + \frac{mv_{2x}^2}{L} + \frac{mv_{3x}^2}{L} + \dots + \frac{mv_{nx}^2}{L} \\ F_x &= \frac{m(v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{nx}^2)}{L} \end{aligned}$$

As pressure is normal force per unit area, hence pressure P on the face perpendicular to x-axis is:

$$\begin{aligned} P &= \frac{F_x}{A} \\ P &= \frac{F_x}{L^2} \\ P &= \frac{m}{L^2} \frac{(v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{nx}^2)}{L} \\ P &= \frac{m}{L^3} (v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{nx}^2) \dots \dots \dots (4) \end{aligned}$$

The number of molecules in unit volume n_v is $\frac{N}{L^3}$. Where N is the total number of molecules. Therefore:

$$L^3 = \frac{m}{n_v}$$

Substituting this value in eq: (4). Therefore

$$P = m \cdot n_v \left(\frac{v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{nx}^2}{N} \right) \dots \dots \dots (5)$$

Where:

$m \cdot n_v$ = Mass per unit volume which we call density ρ and

$$\frac{v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{nx}^2}{N} = \text{Average value } v_x^2$$

For all the molecules in the container, we call this average square velocity v_x^2 . The square root of v_x^2 is referred to as V_{rms} , therefore eq: (5) can be written as:

$$P = \rho v_x^2 \dots \dots \dots (6)$$

The terms v_x^2 is only one component of the total velocity.

The velocity ' v ' of a molecule may be divided into three rectangular components v_x, v_y and v_z , such that root mean square velocity ' v ' of the molecules is

$$\text{Since } v^2 = v_x^2 + v_y^2 + v_z^2$$

$$\text{Where } v_x = v_y = v_z.$$

Since molecules move in all directions with equal speeds i.e.

$$v^2 = v_x^2 + v_x^2 + v_x^2$$

$$v^2 = 3 v_x^2$$

$$v_x^2 = \frac{1}{3} v^2$$

By putting in equation (6), we get

$$P = \frac{1}{3} \rho v^2$$

PROVE THAT: $K.E \propto T$:

According to kinetic molecular theory of gases, the pressure exerted by gasses molecules is

$$P = \frac{1}{3} \rho v^2$$

$$P = \frac{1}{3} \cdot m \cdot n_V v^2 \dots\dots\dots (1) \quad \text{Since } m \cdot n_V = \rho$$

Since n_V represents the number of molecules per unit volume i.e.

$$n_V = \frac{N}{V}$$

Therefore Eq: (1) can be written as

$$P = \frac{1}{3} \cdot m \cdot \frac{N}{V} v^2$$

$$PV = \frac{1}{3} \cdot m \cdot N v^2 \dots\dots\dots (2)$$

Now we can compare the equation (2) with this ideal gas equation.

$$PV = nRT.$$

The left-hand sides are the same. So, the two right-hand sides must also be equal.

$$\frac{1}{3} \cdot m \cdot N v^2 = nRT \dots\dots\dots (3)$$

Substituting $n = \frac{N}{N_A}$ and multiplying both sides by $\frac{3}{2}$, we obtain the relation.

$$\frac{3}{2} \times \frac{1}{3} \cdot m \cdot N v^2 = \frac{3}{2} \frac{N}{N_A} RT$$

$$\frac{1}{2} \cdot m \cdot N v^2 = \frac{3}{2} \frac{R}{N_A} NT$$

Divided both sides by Nm we get.

$$\frac{1}{2} \cdot m \cdot v^2 = \frac{3}{2} \left(\frac{R}{N_A} \right) T$$

Where $\frac{R}{N_A} = k$; Boltzmann constant, equal to 1.38×10^{-23} J/Molecule. K

$$\text{Therefore } \frac{1}{2} \cdot m \cdot v^2 = \frac{3}{2} KT$$

$$(K.E)_{av} = \frac{3}{2} KT$$

$$\text{Hence } (K.E)_{av} \propto T$$

Conclusion:

The mean translational Kinetic energy of a molecule of an ideal gas is proportional to the absolute temperature.

WORKED EXAMPLE: 15.41

The atoms in a gas have mean translational (K.E) equal to 5.0×10^{-21} J. Calculate the temperature of the gas in K and °C.

SELF-ASSESSMENT QUESTIONS:

1. Describe the motion of molecules in a gas according to kinetic molecular theory.
2. What is the relationship between temperature and molecular motion in a gas?

SUMMARY

- Temperature is a measure of the average kinetic energy of particles in a substance.
- The triple point of water is the temperature (0.01 °C) and pressure (611.73 Pascal's) at which water coexists in all three phases (solid, liquid, and gas).
- The general gas law states that $PV = nRT$, relating pressure, volume, number of moles, and temperature of a gas.
- Boyle's Law: $P_1V_1 = P_2V_2$ (pressure and volume are inversely proportional at constant temperature).
- $V_1/T_1 = V_2/T_2$ (volume and temperature are directly proportional at constant pressure).
- $V = nRT/P$ (volume and number of moles are directly proportional at constant temperature and pressure).
- KTG (Kinetic Theory of Gases) Postulates:
 - Gases consist of tiny particles (molecules) that are in constant motion.
 - The molecules are very small compared to the distance between them.
 - The molecules are in constant random motion.
 - The molecules collide with each other and the container walls.
- Molecular movement and pressure: The pressure exerted by a gas is caused by the collisions of molecules with the container walls, and is related to the temperature and volume of the gas by the equation $P = (2/3) nkT$.
- The kinetic energy of molecules is directly proportional to the temperature of the gas, and is described by the equation $KE = (3/2) nkT$.

TEXTBOOK EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS

Tick Mark the correct answer:

1. The relationship between temperature and average kinetic energy of particles in a gas is:
(a) Temperature is inversely proportional to average kinetic energy
(b) Temperature is directly proportional to average kinetic energy
(c) Temperature is independent of average kinetic energy
(d) Temperature is proportional to the square of average kinetic energy
2. Standard conditions of temperature and pressure (STP) refer to a gas at:
(a) 0 °C and 1 atm
(b) 20°C and 1 atm
(c) 25 °C and 1 atm
(d) 30°C and 101.3 kPa (1 atm)
3. If the temperature is kept constant and the volume of a gas is doubled, then pressure of a gas is:
(a) Reduced to $\frac{1}{2}$ of the original value. (b) Doubled
(c) Reduced to $\frac{1}{4}$ of the original value. (d) Quadrupled
4. The Avogadro's number is the number of molecules in:
(a) One mole of a substance
(b) One kg of a substance
(c) One m³ of a gas
(d) One kilogram of hydrogen gas
5. Mean translational K.E. per molecule of an ideal at temperature T is:
(a) $\frac{3}{2} kT$ (b) $\frac{1}{2} kT$ (c) $\frac{2}{3} kT$ (d) kT^4
6. The normal human body temperature is:
(a) 98.6°F (37°C) (b) 99.6°F (37.4°C) (c) 100.4°F (38°C) (d) 101°F (38.3°C)

7. The pressure P , the density Q and the average speed of molecules of an ideal gas are related by the equation.
(a) $p = \frac{2}{3} m \ell^2$ (b) $p = \frac{1}{3} m \ell^2$ (c) $p = \frac{1}{3} \overline{v^2}$ (d) $p = \frac{2}{3} \ell^2$
8. In air at S.T.P, the average speed of the:
(a) Oxygen molecules is greater than Nitrogen molecules
(b) Nitrogen molecules is greater than Oxygen molecules
(c) Oxygen molecules is approximately equal to Nitrogen molecules
(d) Helium atoms is greater than both Oxygen and Nitrogen molecules
9. If the absolute temperature of a gas is increased 3 times, the rms velocity of the molecules will be:
(a) 3 times (b) 9 times (c) $\sqrt{3}$ times (d) $\frac{1}{2}$ times
10. A gas is enclosed in an isolated container which is placed on a fast-moving train uniformly. The temperature of the gas:
(a) Increases due to the motion of the train
(b) Decreases due to the motion of the train
(c) Remains constant,
(d) Fluctuates, depending on the train's speed and direction

SECTION- B: SHORT QUESTIONS:

- Why the earth is not in thermal equilibrium with the sun?
- Describe the relationship between temperature and kinetic energy of molecules.
- It is observed that when mercury in glass thermometer is put in a flame, the column of mercury first descends and then rises. Explain.
- What is standard temperature, pressure?
- A thermometer is placed in direct sun light. What will it read the temperature?
- The pressure in a gas cylinder containing hydrogen will leak more quickly than if it is containing oxygen. Why?
- When a sealed thermos bottle full of coffee is shaken, what are the changes occur?
- How does the Kinetic theory account for the following observed facts?
(a) A gas exerts pressure
(b) The pressure of a gas depends upon its temperature.
- Calculate the average speed of an air molecule at room temperature (20°C) and compare it to the speed of sound in air (330 m/s).

SECTION- C: DETAILED QUESTIONS:

- What is temperature? Explain the scales of temperature in detail.
- Define and explain Boyle's law, Charles's and Avogadro's law.
- Derive general gas law by making use of gas laws.
- Describe the molecular movement causes the pressure exerted by gas, derive pressure equation.
- Interpret mathematically that temperature is a measure of average translational K.E of the molecules of a gas.

SECTION- D: NUMERICAL

- The freezing point of mercury is -39°C . Convert it into $^\circ\text{F}$ and the comfort level temperature of 20° into Kelvin. (Ans. -38.2°F , 293K)
- The boiling point of liquid nitrogen is -321°F . Change it into equivalent Kelvin temperature. (Ans. 77K)

3. Calculate the volume occupied by a gram-mole of a gas at 0 °C and a pressure of 1.0 atmosphere. (Ans. 22.4 liters/mole)
4. An air storage tank whose volume is 112 liters contain 3kg of air at a pressure of 18 atmospheres. How much air would have to be forced into the tank to increase the pressure to 21 atmospheres, assuming no change in temperature? (Ans. 0.5kg)
5. A balloon contains 0.04m³ of air at a pressure of 120KPa. Calculate the pressure required to reduce its volume to 0.025 m³ at constant temperature. (Ans. 1.9 x 10⁵ Pa)
6. The molar mass of nitrogen gas N₂ is 28gmoi⁻¹. For 100g of nitrogen, calculate.
(a) The number of moles. (Ans. (a) 3.57 mole)
(b) The volume occupied at room temperature (20°C) and pressure of 1.01 x 10⁵ Pa. (Ans. (b) 0.086m³ or 86dm³)
7. A sample of a gas contains 3.0 x 10²⁴ atoms. Calculate the volume of the gas at a temperature of 300K and a pressure of 120K Pa. (Ans. 0.104m³)
8. Calculate the root mean square speed of hydrogen molecules at 0 °C and 1.0 atm pressure. Assuming hydrogen to be an ideal gas. The density of hydrogen is 8.99 x 10⁻² kg/m³ (Ans. 1835.86ms⁻¹)
9. Calculate the root mean square speed of hydrogen molecule at 500K (mass of proton = 1.67 x 10⁻²⁷kg and K = 1.38 x 10⁻²³J/molecule.-K) (Ans. 2489.49ms⁻¹)
10. (a) Determine the average value of the Kinetic energy of the particles of an ideal gas at 10°C and at 40°C. (Ans. 5.86 x 10⁻²¹ J, 6.48 x 10⁻²¹ J)
(b) What is the Kinetic energy per mole of an ideal gas at these temperatures? (Ans. 3526.57J, 3901J)

ASSIGNMENT: 15

MULTIPLE CHOICE QUESTIONS

HEAT:

1. Heat Energy cannot be measured in: (2018, 14, 09)
(a) Joule (b) B.T.U. (c) **Kelvin** (d) Calorie
2. The sum of all the energies of molecules in a substance is called: (2013)
(a) **Heat energy** (b) Temperature (c) Kinetic energy (d) Potential energy
3. In C.G.S. system one calorie of heat is equal to: (2006)
(a) 11.184J (b) 2.184J (c) 3.184J (d) **4.184J**
4. One cubic meter is equal to: (2003,06)
(a) **10⁶ cm³** (b) 10² cm³ (c) 10³ cm³ (d) 10⁻³ cm³
5. The S.I unit of heat is: (2002E)
(a) **Joule** (b) Calorie (c) Centigrade (d) Fahrenheit

TEMPERATURE:

6. The value of absolute zero on the Fahrenheit scale is: (2022)
(a) 491.4°F (b) -459.4°F (c) -523.4°F (d) 459.4°F
7. The average internal energy of an ideal gas is called: (2014)
(a) Pressure (b) Volume (c) **Temperature** (d) Heat
8. A temperature of 50°C is equal to: (2016)
(a) 105°F (b) 60°F (c) **122°F** (d) 120°F
9. Absolute zero corresponds to this temperature on Fahrenheit scale (2012)
(a) 32 (b) -180 (c) **-460** (d) 212
10. On Fahrenheit scale the temperature of 50°C will be: (2010)

- (a) 40°F (b) 10°F (c) **122°F** (d) 105°F
11. Fahrenheit and Celsius scales of temperature coincide at: (2007)
(a) 0° (b) 237° (c) -273° (d) **-40°**
12. In Celsius scale 1°C in magnitude is equal to (2006)
(a) 32°F (b) 16°F (c) 0°F (d) **1.8°F**
13. The absolute temperature corresponding to 212°F is: (2005)
(a) 485 K (b) **373 K** (c) 161 K (d) 100 K
14. 273 Kelvin corresponds to (2003)
(a) 273°C (b) -32°F (c) 0°C (d) -273°C
15. Zero on the Celsius scale is equal to: (2002E)
(a) **273K** (b) 32K (c) 100K (d) 212K
16. The temperature on Fahrenheit scale corresponding to absolute zero is: (2002M)
(a) 32°F (b) -180°F (c) **-460°F** (d) 212°F
17. The temperature at which centigrade scale is equal to Fahrenheit scale is: (1993)
(a) 0 (b) -32 (c) **-40** (d) 100

GAS LAWS:

18. For an ideal gas at constant temperature, the graph between V and 1/p is: (2021)
(a) Hyperbola (b) Parabola (c) **Straight line** (d) Circle
19. A container is filled with a sample of an ideal gas at a pressure of 1.5 atm. The gas is compressed isothermally to one fourth of its original volume. Its new pressure will be: (2019)
(a) 2 atm (b) 4 atm (c) 6 atm (d) 9 atm
20. The volume of a given gas at constant pressure becomes zero at: (2007)
(a) 273 K (b) 273°C (c) -273 K (d) **-273°C**
21. If the volume of a given mass of a gas is doubled without changing its temperature, the pressure of the gas is. (2008)
(a) **reduced to ½ of the initial value** (b) the same as the initial value
(c) reduced to ¼ of the initial value (d) double of the initial value
22. If the pressure of a gas is doubled, keeping the temperature constant, the volume of the gas is: (2004)
(a) reduced to one-fourth (b) doubled
(c) reduced to one-half of the original volume (d) increased four times
23. According to Charles's Law: (2003)
(a) VT = constant when the pressure is constant
(b) V/T = Constant when the pressure is constant.
(c) PV = constant when the temperature is constant
(d) None of these.
24. Choose the correct answer for each from the given option: (2002E)
(a) The product of P and T is constant if the volume is constant.
(b) The ratio of P and V is constant if the temperature is constant
(c) The ratio of P and V is constant if the temperature is constant
(d) The product of V and T is constant if the pressure is kept constant
25. The pressure exerted by a column of mercury 76 cm high at 0°C is called: (2002M)
(a) 1 liter (b) 1 cm³ (c) **1 atmosphere** (d) 1 N/m²
26. Real gas molecules do not strictly obey gas laws at: (2001)

(a) High pressure and low temperature

(b) Low pressure and high temperature

(c) High pressure and high temperature

(d) None of the above,

K.T.G:

27. The average translational kinetic energy per mole of molecules of a gas is: **(2019, 18, 15, 13, 11, 08)**
 (a) $\frac{3}{2} KT$ (b) $\frac{2}{3} KT$ (c) $\frac{3}{2} RT$ (d) $\frac{2}{3} RT$
28. According to Kinetic Theory of gases the absolute temperature of a perfect gas is: **(2007)**
 (a) Inversely proportional to the K.E. of the molecules
 (b) Independent of kinetic energy of the molecules
 (c) Equal to the kinetic energy of the molecules
 (d) Directly proportional to the average translational kinetic energy of the molecules.
29. Boltzmann Constant is equal to: **(2003)**
 (a) RN_A (b) R/N_A (c) N_A/R (d) $\frac{\sqrt{R}}{N_A}$
30. The value of Boltzmann's constant is
 (a) $3.85 \times 10^{-23} \text{ Jk}^{-1}$ (b) $2.185 \times 10^{-12} \text{ Jk}^{-1}$ (c) $1.62 \times 10^{-22} \text{ Jk}^{-1}$ (d) $1.38 \times 10^{-23} \text{ Jk}^{-1}$
31. The universal gas constant per molecule is called **(2002M)**
 (a) Planck's constant (b) Stefan's constant
 (c) Avogadro's constant (d) **Boltzmann's constant**
32. The K.E. of the molecules of an ideal gas at absolute zero temperature will be: **(2001)**
 (a) Infinite (b) **Zero** (c) Very high (d) below zero
33. The Pressure exerted by the gas molecules on the walls of a vessel increases if the: **(2001)**
 (a) Temperature of the gas decreases
 (b) Velocity of the molecules of the gas decreases
 (c) **Collision of the molecules of the gas with its walls increases.**
 (d) Neither heat nor work is inter-convertible
34. According to Kinetic Theory of gases, the root mean square speed of the gas molecules is directly proportional to: **(1993)**
 (a) **T** (b) T_1 (c) T_2 (d) T_4

SUBJECTIVE QUESTIONS**HEAT & TEMPERATURE:**

1. Differentiate between Heat and Temperature in the light of the molecular concept of matter. **(1993)**

GAS LAWS:

2. State perfect gas laws and show how they may be combined to give the equation of state of an ideal gas. **(2006, 1993)**

K.T.G:

3. State the basic assumptions of Kinetic Molecular Theory of gases. **(2007, 98, 96)**
4. Derive pressure formula of an ideal gas using kinetic molecular theory. **(2022)**
5. Derive a relation for the pressure of an ideal gas in terms of its density and mean square velocity. **(2017, 12, 10, 09, 07, 03M, 96)**
6. On the basis of kinetic theory of gases, show that $P = \frac{1}{3} \rho \overline{v^2}$. **(2014)**

7. On the basis KMT of gases, show that: $\frac{1}{2}mv^2 = \frac{3}{2}KT$. (2015)
8. Show that the absolute temperatures of an ideal gas is directly proportional to the average translational kinetic energy of the molecules. (2010, 09, 07, 06, 02M, 98)
9. Using the relation $P = \frac{1}{3} \rho v^2$, deduce the expressions of the Boyle's Law and Charles Law (2008, 07, 02)
10. Verify Boyle's Law and Charles's Law on the basis of kinetic molecular theory. (2000)

PROBLEMS

THERMOMETER:

1. A Celsius thermometer in a laboratory reads the surrounding temperature as 30°C: What is the temperature in the Fahrenheit scale and in absolute scale? (Ans: 86°F, 303K) (2006)
2. The normal body temperature is 98.4 F. What is this temperature on Celsius scale? Ans: 36.88°C
3. At what temperature do the Fahrenheit and Celsius scales coincide? Ans: 40°

GAS LAWS:

4. A scientist stores 22 gm of gas in a tank at 1200 atmospheres. Overnight the tank develops slight leakage, and the pressure drops to 950 atmospheres. Calculate the mass of the gas escaped. (Ans: 4.6 gm) (2009)
5. An air-storage tank whose volume is 110 liters contains 2 kg of air at a pressure of 15 atmospheres. How much air would have to be forced into the tank to increase the pressure to 18 atmospheres, assuming no change in temperature? (Ans: 0.4 Kg) (2003M)
6. Calculate the volume occupied by a gram mole of a gas at 10°C and pressure of one atmosphere. (Given $R = 8.313 \text{ J mol}^{-1} \text{ K}^{-1}$, 1 atmosphere = $1.01 \times 10^5 \text{ N/m}^2$) (2000)
7. Calculate the volume occupied by a gram-mole of a gas at 0°C and a pressure of 1 atmosphere.

K.T.G:

8. Calculate the temperature at which the root mean square speed of hydrogen molecules is 3300 m/s. Give your answer in degree Celsius ($m = 3.32 \times 10^{-27} \text{ kg}$). (2017)
9. Calculate the root mean square speed of Oxygen molecule at 800 K. Its molar mass is 32 gm and universal gas constant $R = 8.31 \text{ J/mole-K}$. (2014)
10. Calculate the density of hydrogen gas, considering it to be an ideal gas, when the root mean square velocity of hydrogen molecules is 1850 m/sec at 0°C and 1 atmospheric pressure. (Ans: 0.08853 Kg/ m³) (2007, 2004)
11. Find the root mean square of a hydrogen molecule at 7°C. Take the mass of a hydrogen molecule to be $3.32 \times 10^{-27} \text{ kg}$ and Boltzmann's constant = $1.38 \times 10^{-23} \text{ joules/}^\circ\text{K}$. (Ans: 1868.6 m/s) (2002M, 98)
12. Find the root mean square velocity of the nitrogen molecules at 27°C. Given the mass of nitrogen molecule to be $4.67 \times 10^{-26} \text{ kg}$. $K = 1.38 \times 10^{-23} \text{ joule/K}$. (Ans: 515.7 m/s) (1996)
13. Calculate the V_{rms} of hydrogen molecules at 0.00°C and 1.00 atm pressures, assuming hydrogen to be an ideal gas. Under these conditions hydrogen has a density ρ of $8.99 \times 10^{-2} \text{ kg/m}^3$.
14. (a) Determine the average value of the kinetic energy of the particles of an ideal gas at 0°C and at 50°C. (Ans: $5.65 \times 10^{21} \text{ J}$, $6.68 \times 10^{21} \text{ J}$)
- (b) What is the kinetic energy per mole of an ideal gas at these temperatures?

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